



Intellectual and developmental disability and COVID-19 case-fatality trends: TriNetX analysis

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Brief Report

Intellectual and developmental disability and COVID-19 case-fatality trends: TriNetX analysis



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ABSTRACT

Background: Despite possibly higher risk of severe outcomes from COVID-19 among people with intellectual and developmental disabilities (IDD), there has been limited reporting of COVID-19 trends for this population.

Objective: To compare COVID-19 trends among people with and without IDD, overall and stratified by age.

Methods: Data from the TriNetX COVID-19 Research Network platform was used to identify COVID-19 patients. Analysis focused on trends in comorbidities, number of cases, number of deaths, and case-fatality rate among patients with and without IDD who had a positive diagnosis for COVID-19 through May 14, 2020.

Results: People with IDD had higher prevalence of specific comorbidities associated with poorer COVID-19 outcomes. Distinct age-related differences in COVID-19 trends were present among those with IDD, with a higher concentration of COVID-19 cases at younger ages. In addition, while the overall case-fatality rate was similar for those with IDD (5.1%) and without IDD (5.4%), these rates differed by age: ages ≤ 17 – IDD 1.6%, without IDD $<0.01\%$; ages 18–74 – IDD 4.5%, without IDD 2.7%; ages ≥ 75 – IDD 21.1%, without IDD, 20.7%.

Conclusions: Though of concern for all individuals, COVID-19 appears to present a greater risk to people with IDD, especially at younger ages. Future research should seek to document COVID-19 trends among people with IDD, with particular attention to age related trends.

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Background

- Higher risk of severe outcomes from COVID-19 among people with IDD.
- Limited reporting of COVID-19 trends for this population.
- IDD includes intellectual disability, cerebral palsy, Down syndrome, etc.

#note : IDD=Intellectual and Developmental Disability

Objective

- Compare COVID-19 trends among people with and without IDD.
- Analysis stratified by age.

Methods

- Data from TriNetX COVID-19 Research Network platform.
 - Included data from 42 healthcare organizations.
 - Patients categorized based on IDD status.
 - Comparison of comorbidities, number of cases, number of deaths, and case-fatality rates. 共病、病例數、死亡數和病死率
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Results

Comorbidities 合併症

- Higher prevalence of specific comorbidities among IDD patients.
- Comorbidities include respiratory, endocrine, and circulatory system diseases. 共病包括呼吸系統疾病、內分泌疾病和循環系統疾病

Case Distribution 病例分佈

- Higher concentration of COVID-19 cases at younger ages among IDD patients.
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Results

Case-Fatality Rates 病死率

- Overall case-fatality rate similar for IDD (5.1%) and non-IDD (5.4%) patients. IDD患者 (5.1%) 和無IDD患者 (5.4%) 的總體病死率相似
 - Age-related differences in case-fatality rates:
 - Ages ≤ 17 : IDD 1.6%, without IDD $< 0.01\%$
 - Ages 18-74: IDD 4.5%, without IDD 2.7%
 - Ages ≥ 75 : IDD 21.1%, without IDD 20.7%
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Discussion



- Higher comorbidity rates among IDD patients may lead to higher COVID-19 case-fatality rates.
 - Age-related trends show higher case-fatality rates at younger ages for IDD patients.
 - Need for more research to understand COVID-19 trends among IDD patients.
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Conclusion

- IDD patients have higher prevalence of comorbidities associated with poorer COVID-19 outcomes.
 - COVID-19 presents a greater risk to IDD patients, especially at younger ages.
 - Future research should focus on documenting COVID-19 trends among IDD patients, with attention to age-related trends.
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THANKYOU